Database Requirements Summary

Production AWS RDS Deployment Specification for Sterling Checkout

1. Executive Summary

Sterling Checkout is a customer-facing, high-performance SaaS platform built to orchestrate and manage high-volume checkout sequences, transaction payloads, and merchant orders on a global scale, serving markets in North America and Europe. To maintain seamless transaction lifecycle integrity, structural resilience, and absolute security, the production data layer must be deployed onto a fully managed AWS RDS instances architecture. This blueprint defines the exhaustive architectural requirements, network parameters, and security bounds necessary to support this enterprise-grade transactional engine.

2. Database Engine & Topology Specifications

The relational model of transactional processing (order placement, ledger posting, and concurrent payment state updates) requires absolute ACID compliance, high throughput capacity, and deterministic query performance. The chosen technology is a fully managed AWS RDS PostgreSQL or Amazon Aurora PostgreSQL implementation.

Architectural Dimension	Production Target Specification
Database Engine Family	Amazon Aurora PostgreSQL (or AWS RDS PostgreSQL 15+ equivalent)
Deployment Model	Multi-AZ (Availability Zone) Replication with automated synchronous failover topology
Target Environment	Production Tier (Sterling Checkout Core Application Layer Context)
Primary Regional Footprint	Primary cluster serving NA/EU merchant routing demands with optimized low-latency data structures

3. Transactional Workload Profile

The application environment demonstrates a highly demanding transaction profile characterized by continuous baseline operations punctuated by massive, unpredictable volume bursts corresponding to merchant marketing spikes, holiday shopping cycles, and flash sales. The architecture is optimized against three fundamental dimensions:

• **High Transactions Per Second (TPS):** Sub-second processing pipelines for order write paths, checkout mutations, and synchronized payment ledger balance records.

• **Read/Write Balance:** Write-heavy ingest pipelines for real-time order creations paired with intense, concurrent, and bursty read lookup patterns originated by merchant administration dashboards, reporting microservices, and client checkouts.

• **Low-Latency UI/UX Bounds:** Stringent performance targets dictating sub-15ms database execution response windows for critical write paths to protect conversion metrics and ensure fluid checkout flows.

4. Network Isolation & Perimeter Architecture

Per network security architecture best practices, the database tier is completely air-gapped and decoupled from public networks. The deployment completely denies any public internet path or wildcard ingress routes.

Network Layer Component	Structural Architecture & Configuration Boundary
Virtual Private Cloud (VPC)	Isolated Corporate Application VPC defined via broad block **10.0.0.0/16**
Availability Zone A Placement	Private Subnet **10.0.1.0/24** (Strictly non-public; no Internet Gateway routing)
Availability Zone B Placement	Private Subnet **10.0.2.0/24** (Strictly non-public; target for Multi-AZ sync replication)
Public Ingress Accessibility	**DISABLED** (Publicly Accessible flag set explicitly to false; zero direct public paths)

5. Data Protection, Security Controls & Access Governance

To satisfy international payment industry compliance baselines and safeguard merchant intelligence, data security follows a strict defense-in-depth model across structural segments:

• **Inbound Access Control:** Strict network security group access bounds. Ingress connections to port 5432 are explicitly limited to authorized application components, namely the *Checkout Service*, *Order Service*, *Payment Service*, and the *Reporting Service*.

• **Encryption Mechanics:** Cryptographic enforcement is active across both storage and transmission vectors. Data-at-rest is encrypted natively utilizing the **AWS KMS (Key Management Service) AES-256** cipher suite. Data-in-transit enforces mandatory **TLS 1.2+** encapsulation for every inbound connection payload, discarding cleartext attempts.

• **Administrative Infrastructure:** Database administration, maintenance operations, and schema patching are mediated exclusively via an isolated **AWS Bastion Host** requiring mandatory multi-factor authentication (MFA). Direct developer access is blocked. All administrative actions undergo time-bounded session token distribution (Just-In-Time) and complete end-to-end centralized audit logging.

• **Privilege Model:** Elimination of generalized root accounts. Application microservices connect using individual, unique service-account database users provisioned under strict least-privilege role boundaries.

6. High Availability & Scalable Storage Objectives

To protect against physical infrastructure failures and volume scaling bottlenecks, the instance is engineered with automated operational primitives:

• **High Availability:** The Multi-AZ synchronous storage replication topology ensures a zero-RPO failover pattern. In the event of a primary compute failure, AWS RDS handles automated DNS routing redirection to the standby replica, preserving client uptime bounds.

• **Storage Growth Architecture:** Storage configurations run on scalable high-performance volumes with storage auto-scaling activated, allowing seamless runtime scale-up as data footprints expand from transaction accumulation.

• **Managed Lifecycles:** Automated backup routines operate within designated daily low-traffic windows, generating transaction logs and point-in-time recovery maps to satisfy strict operational durability and platform SLAs.